MILITARY SPECIFICATION

PANEL, POWER

(FOR TEST STAND MK 7 MOD 0)

1. SCOPE

1.1 Scope. - This specification states the requirements for the procurement of a power panel, BuOrd LD 290974, for use with Test Stand Mk 7 Mod 0 as standard torpedo workshop equipment. The power panel operates under circuits of 24 volts d.c. and 115 volts a.c. 3 phase 400 cycle.

2. APPLICABLE DOCUMENTS

2.1 The following specifications, standards, drawings and publications, or such portions thereof as may be designated herein, of the issue in effect on date of invitation for bids, form a part of this specification.

SPECIFICATIONS

MILITARY

MIL-P-116	Preservation, Methods of
JAN-P-658	Packaging and packing of electrical equipment and spare parts (electronic, electrical, and electro-mechanical)
MIL-T-17113 (Ships)	Tests; Shock, Vibration, and Inclination (for electronic equipment) General Specification
MIL-T-18404 (NOrd)	Torpedoes, Environmental Requirements; General Specification for

FSC 4925

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STANDARDS

MILITARY

MIL-STD-105	Sampling Procedure and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage

NAVY DEPARTMENT

BUREAU OF ORDNANCE

NAVORD OSTD-2	Method of Marking Ordnance Material

DRAWINGS

NAVY DEPARTMENT

BUORD LD 290974	POWER PANEL
BUORD LD 1362354	TEST STAND MK 7 MOD 0
BUORD DWG 978611	AFTERBODY AND TAIL CONE SECTIONS

PUBLICATIONS

BUREAU OF ORDNANCE

OP 400	General Instructions for the Design, Manufacture and Inspection of Naval Ordnance Equipment
NAVORD OS 6341	Miscellaneous General Ordnance Design Requirements

(Copies of specifications, standards, drawings and publications requested by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. - The following documents form a part of this specification. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

NATIONAL BUREAU OF STANDARDS PUBLICATIONS

Handbook H28 and Supplements	Screw Thread Standards for Federal Services

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(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington 25, D. C.)

3. REQUIREMENTS

3.1 Description. - The power panel, as a sub-assembly mounted on Test Stand Mk 7 Mod 0, BuOrd Drawing 1362354, is primarily an electrical system for testing and adjusting the electrical systems of an afterbody under test. The power panel is used in conjunction with a 24 volt d.c. power supply and a 115 volt 3 phase 400 cycle power supply for testing and adjusting the electrical circuits of an afterbody, BuOrd Drawing 978611, in a torpedo workshop.

3.2 General requirements

3.2.1 Pre-production samples. - Unless otherwise specified in the contract or order, pre-production samples of the power panel shall be manufactured using the methods and procedures proposed for the production lot. These samples will be tested as specified in section 4 herein and are for the purpose of determining that the production item meets the requirements of this specification, and the design. See paragraph 4.1.2.

3.2.2 Materials. - Unless otherwise specified in the contract, requisition, or order, all materials and purchased parts entering into the final assembly of the power panel shall conform strictly to the detailed requirements indicated on the drawings and specifications listed in BUORD LD 290974.

3.2.3 Finishes, coatings, construction, dimensions and tolerances. - Finishes, coatings, construction, dimensions and tolerances for the power panel shall be in strict accordance with the applicable drawings, specifications and publications listed in BUORD LD 290974.

3.2.4 Moisture, fungus, and corrosion resistance. - The power panel shall be rendered moisture, fungus, and corrosion resistant in accordance with the detailed requirements indicated on the drawings and specifications in BUORD LD 290974. When the drawings and specifications do not specify these requirements the panel shall meet the requirements of NAVORD OS 6341.

3.2.5 Marking. - Marking of assemblies, sub-assemblies and parts shall conform to the applicable drawings listed in BUORD LD 290974 and to NAVORD OSTD-2.

3.2.6 Interchangeability. - Component parts shall be interchangeable without interference, binding, or misalignment, and without resorting to selection or selective fitting.

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3.2.7 Gages. - The contractor shall provide himself with whatever gages are necessary and adequate to insure that the material to which this specification applies will meet the dimensional requirements shown on the applicable drawings.

3.2.8 Threads. - Unless otherwise specified, all threads shall be in accordance with National Bureau of Standards Handbook H28 and supplements thereto. The class or fit for threads shall be as specified on the applicable drawings, and the NOT-GO gaging practice shall be as set forth in the Supplement to Handbook H28.

3.2.9 Requests for deviations. - Requests for deviation from this specification, applicable drawings, specifications, publications, materials and processes specified herein shall be as set forth in OP 400.

3.2.10 Conflicting requirements. - Conflicting requirements arising between this specification and any specifications, publications, or drawings listed herein shall be referred in writing to the procuring agency or appointed agent for interpretation and clarification.

3.3 Definitions. - Definitions applicable to the environmental requirements of this specification are as set forth in MIL-T-18404.

3.4 Performance and product characteristics. - The power panel shall meet the following performance and product characteristics:

3.4.1 Insulation resistance. - There shall be a resistance of at least 3 megohms at specified insulation test points when the power panel is checked as provided in 4.5.4.1.

3.4.2 Circuit continuity and resistance. - Designated circuits of the power panel shall be continuous and resistance between specified terminals shall be less than 0.5 ohms when checked as provided in 4.5.4.2 and 4.5.4.3.

3.4.3 Meter accuracy. - Meters of the power panel shall be accurate within the prescribed limits when tested in accordance with 4.5.4.4.

3.5 Environmental conditions. - The power panel shall meet the performance and product characteristics of this specification after or while being subjected to the following requirements, when tested as defined under section 4 of this specification.

3.5.1 Vibration. - The power panel shall be subjected to the vibration tests specified in MIL-T-17113 and thereafter show no damage or impaired performance or product characteristics.

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3.5.2 Shock. - The power panel shall be subjected to type I shock tests as specified in MIL-T-17113 and thereafter show no damage or impaired performance when tested in accordance with section 4 of this specification.

3.5.3 Inclination. - The power panel shall be subjected to inclination tests as specified in MIL-T-17113 and thereafter show no damage or impaired performance when tested in accordance with section 4 of this specification.

3.5.4 Temperature and humidity

3.5.4.1 Temperature and humidity cycling. - The power panel shall be exposed to temperature and humidity cycling as specified by MIL-T-18404 (NOrd) for "torpedo non-operating" condition and thereafter shall show no damage or impairment in performance or product characteristics.

3.5.4.2 Low and high temperature performance. - The power panel shall be exposed to temperature extremes of 0°F and 130°F and while at test temperatures shall be capable of meeting the performance and product characteristic requirements of this specification.

3.5.5 Tropical environment. - The power panel shall be exposed to the tropical environment conditions specified in MIL-T-18404 and thereafter show no damage or impaired performance or product characteristics.

3.5.6 Sunlight. - The power panel shall be exposed to the conditions of sunlight set forth in MIL-T-18404 and thereafter show no damage or impaired performance or product characteristics.

3.5.7 Marine environment. - The power panel shall be exposed to the simulated sea atmosphere test specified in MIL-T-18404 and thereafter shall show no evidence of corrosion or damage, or impaired performance and product characteristics.

3.5.8 Storage life. - The power panel shall be capable of being stored for a period of ten (10) years when preserved and packaged in accordance with section 5 of this specification and five (5) years in an unpackaged condition, under the following temperature and humidity conditions:

3.5.8.1 Temperature: -65°F to +160°F.

3.5.8.2 Humidity: Up to 95% relative maximum.

3.6 Clean-up. - Prior to and after final assembly all parts, components and the assembly shall be thoroughly cleaned of loose, spattered or excess solder, metal chips, and other foreign matter. Burrs and sharp edges as well as rosin flash shall be removed.

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3.7 Workmanship. - The equipment, including all parts and accessories, shall be constructed and finished in a manner to assure compliance with all requirements of this specification. Particular attention shall be paid to neatness and thoroughness of soldering, wiring, marking of parts and assemblies, plating, painting, machine screw assemblage, brazing and freedom of parts from burrs and sharp edges.

4. QUALITY ASSURANCE PROVISIONS AND TEST REQUIREMENTS

4.1 Government inspection. - The Government inspector will make such inspections as are necessary to determine that all components and assemblies are in accordance with the requirements of the contract, pertinent drawings and specifications. Unless otherwise specified, these inspections will be conducted in accordance with instructions contained in OP 400.

4.1.1 Sampling. - Unless otherwise specified, and when applicable, the sampling plans and procedures used by the Government inspector in the determination of the acceptability of products submitted by a supplier for Government inspection, shall be in accordance with the provisions set forth in MIL-STD-105.

4.1.2 Pre-production sample. - Unless otherwise specified for each contract or order, a representative sample of three(3) power panels shall be manufactured using the methods and procedures proposed for the production lot. These shall be submitted to the Government inspector for pre-production tests at an activity designated in the contract or by the Bureau of Ordnance, for purposes of determining compliance with the requirements of the contract, specifications and drawings. Further production of the item by the contractor prior to approval of the contracting activity or the completion of tests on the pre-production samples shall be at the contractor's risk. Pre-production samples accepted will be applied as part of the quantity specified by the contract or order.

4.1.3 Lot size. - As applied to Government inspection of units of product, the term "lot" shall mean "inspection lot", i.e., a collection of units of product submitted by a supplier for Government inspection. Unless otherwise specified, the number of units of product in "inspection lots" shall be as determined by the Government inspector and may differ from the quantity designated in the contract or order as a lot for production, shipment, or other purpose.

4.1.4 Periodic production samples. - From each succeeding lot or grand lot of one hundred (100) or less, a random sample of three(3) power panels will be selected by the Government inspector for submission to a Bureau of Ordnance assigned activity to determine compliance of the samples

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with the requirements of the contract, specification, and drawings. The power panel shall be considered satisfactory if it meets all specifications and pertinent drawing requirements, and withstands the tests as specified in 4.5 and 4.6.

4.2 Contractor's inspection. - Contractor's inspection shall be conducted in accordance with provisions of "Inspection by Contractor" as contained in OP 400.

4.3 Classification of tests. - The inspection and testing of the power panel shall be classified as follows:

4.3.1 Acceptance tests. - These tests are to be accomplished on power panels being submitted for acceptance under contract. Acceptance tests shall either be performed by the contractor and witnessed by the Government inspector, or these tests may be conducted by the Government inspector. These tests are detailed in 4.5.

4.3.2 Pre-production and periodic production tests. - Pre-production and periodic production tests are those which are accomplished on samples selected as specified in 4.1.2 and 4.1.4 which are representative of the production of the power panel after the award of the contract to determine that the design and the production meet the requirements of this specification. These are detailed in 4.6. Acceptance shall be based on no defects in the sample. Failure of a sample to comply with these requirements will result in the rejection of the lot or the cessation of production as determined by the procuring activity.

4.4 Test equipment. - The following items of test equipment are required to perform the acceptance tests set forth in this specification.

4.4.1 Insulation test. - A 500 volt megohmmeter with full scale not less than 10 megohms, accurate within ± 10%.

4.4.2 Circuit continuity test. - A source of 24 volt d.c. power ± 2 volts.

4.4.3 Resistance test. - An ohmmeter capable of measuring 0.5 ohms ± 0.1 ohms.

4.4.4 Meter test equipment.

4.4.4.1 Power supply. - 115 volts ± 5 volts, 3 phase, 400 cycles ± 8 cycles, with known phase sequence.

4.4.4.2 Voltmeter. - a.c., scale 0-150 volts, accurate to ± 0.5 volts between 80 and 150 volts.

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4.4.4.3 Frequency meter. - Accurate to ± 0.5 cycles per second in the range 380 to 420 cycles per second.

4.5 Acceptance tests

4.5.1 The contractor or Government inspector shall conduct acceptance tests as specified in 4.3.1 to assure that the power panel is in compliance with the requirements listed under sections 3.2 and 3.4 and paragraphs 3.6 and 3.7.

4.5.2 Test conditions. - Unless otherwise specified, the power panel shall be subjected to acceptance tests under the following conditions:

4.5.2.1 Temperature. - Room ambient 65°F to 95°F (18°C to 35°C).

4.5.2.2 Altitude. - Normal ground

4.5.2.3 Vibration. - None

4.5.2.4 Humidity. - Room ambient to 95% relative maximum.

4.5.3 Test and inspection equipment and facilities. - Unless otherwise specified in the contract or order, the manufacturer shall furnish and maintain all necessary test equipment, facilities and personnel for performing all acceptance tests. The test equipment shall be adequate in quantity and, unless definite requirements are specified, shall be of sufficient accuracy and quality to permit performance of the required tests.

4.5.4 Test procedure

4.5.4.1 Insulation test. - With all switches closed and all plugs disconnected, using the megohmmeter of 4.4.1, measure the insulation resistance between the following points and the panel chassis:

E-1, E-2, J-1-A, J-1-C (with SW-1 on "War"), J-1-D, J-1-G (with SW-1 on "Exercise"), J-1-J, J-1-K, J-1-M, J-1-N, P-1-A, P-1-B, P-1-C, P-1-D, P-2-A, and P-2-B

The resistance reading shall not be less than 3 megohms in any case.

4.5.4.2 Circuit continuity test. - Apply the 24 volt d.c. power source of 4.4.2 to pins A and B of d.c. input connector P-2. Turn all the switches to the "ON" position and check the following:

4.5.4.2.1 Connect a jumper between J-1-N and J-1-A. Lamp I-4 (S-16 "NO") should light. Remove the jumper.

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4.5.4.2.2 Connect a jumper between J-1-D and J-1-J. Lamp I-3 (S-17 "NO") should light. Remove the jumper.

4.5.4.2.3 Connect a jumper between J-1-C and J-1-A with switch SW-1 in the "War" position. Lamp I-4 (S-16 "NO") should light. Remove the jumper.

4.5.4.2.4 Connect a jumper between J-1-G and J-1-A with switch SW-1 in the "Exercise" position. Lamp I-4 (S-16 "NO") should light. Remove the jumper.

4.5.4.2.5 Connect a jumper between motor lead E-1 and J-1-D. Lamp I-3 (S-17 "NO") should light. Remove the jumper.

4.5.4.3 Resistance test. - With all power disconnected and all switches closed, and using the ohmmeter of 4.4.3, measure the resistance between the following points:

P-2-A to J-1-J, J-1-K, J-1-M and E-1
P-2-B to J-1-N, J-1-C (with SW-1 in "War" position),
J-1-G (with SW-1 in "Exercise" position),
and E-2

The resistance shall be less than 0.5 ohms in every case.

4.5.4.4 Meter test. - With the 115 volt power source of 4.4.4.1 connected to plug P-1, with phase one to P-1-B, phase two to P-1-D, and phase three to P-1-A, check the following:

4.5.4.4.1 Lamp I-1 should be on dim, and lamp I-2 should be on bright.

4.5.4.4.2 Connect the a.c. voltmeter of 4.4.4.2 across P-1-D and P-1-C. Meter M-1 should be accurate within ± 2 volts.

4.5.4.4.3 Connect the frequency meter of 4.4.4.3 across P-1-C and P-1-D. Meter M-2 should be accurate within ± 4 cycles per second.

4.6 Pre-production and periodic production tests

4.6.1 Vibration. - The power panel shall be subjected to vibration testing as prescribed by MIL-T-17113 and, at the conclusion of the test, shall meet the performance and other requirements of this specification.

4.6.2 Shock. - The power panel shall be subjected to shock testing as prescribed by MIL-T-17113 for Type I equipment and, at the conclusion of the test, shall meet the performance and other requirements of this specification.

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4.6.3 Inclination. - The power panel shall be subjected to the inclination tests prescribed by MIL-T-17113 and, at the conclusion of the tests, shall meet the performance and other requirements of this specification.

4.6.4 Temperature and humidity tests

4.6.4.1 Temperature and humidity cycling. - The power panel shall be subjected to the temperature and humidity cycling tests prescribed in section 4 of MIL-T-18404 for the "torpedo non-operating" condition and, at the conclusion of the tests, shall meet the performance and other requirements of this specification.

4.6.4.2 Low temperature test. - The power panel shall be placed in a test chamber whose temperature shall be lowered to 0°F ± 5°F and maintained at this temperature for a period of 48 hours ± 5 hours. At the end of this test period, the power panel, while still at the test temperature of 0°F, shall be checked for performance and shall meet the requirements of 3.4.

4.6.4.3 High temperature test. - The power panel shall be placed in a test chamber whose temperature shall be raised to 130°F ± 5°F and maintained at this temperature for a period of 48 hours ± 5 hours. At the end of this test period, the power panel, while still at the test temperature of 130°F, shall be checked for performance and shall meet the requirements of 3.4.

4.6.5 Tropical environment. - The power panel shall be exposed to fungus inoculation as prescribed by section 4 of MIL-T-18404 and, at the conclusion of the tests, shall meet the performance and other requirements of this specification.

4.6.6 Sunlight. - The power panel shall be subjected to the conditions of sunlight exposure defined in section 4 of MIL-T-18404 and, at the conclusion of the test, shall meet the performance and other requirements of this specification.

4.6.7 Marine environment. - The power panel shall be subjected to a simulated sea atmosphere test under the conditions prescribed in section 4 of MIL-T-18404 and, at the conclusion of the test, shall meet the performance and other requirements of this specification.

4.7 Rejections. - Failure of the power panel to meet any of the requirements and tests of this specification shall be considered cause for rejection.

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5. PREPARATION FOR DELIVERY

5.1 Packaging and packing

5.1.1 Domestic shipment. - Unless otherwise specified in the order or contract, packaging and packing for domestic shipment shall be in accordance with approved commercial methods adequate to protect the power panel from injury and to insure safe delivery at destination. The method used shall be commensurate with shipping at the lowest practicable commercial rates.

5.1.2 Overseas shipment. - Unless otherwise specified in the order or contract, the power panel, when prepared for overseas shipment, shall be packaged and packed in accordance with JAN-P-658.

5.1.3 Preparation for storage. - The power panel, when prepared for storage to meet the requirements of 3.5.8, shall be packaged in accordance with method II of MIL-P-116.

5.2 Labeling and marking. - Unless otherwise specified, labeling and marking of all shipping containers shall be in accordance with MIL-STD-129. Each container shall be marked with the serial number of the unit it contains.

6. NOTES

6.1 Intended use. - The power panel is intended for use with Test Stand Mk 7 Mod 0 as standard torpedo workshop equipment.

6.2 Ordering data. - Procurement documents should specify the exact title, number and date of this specification and exceptions to this specification, applicable drawings and other documents.

PATENT NOTICE:

When government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related government procurement operation, the United States Government thereby incurs no responsibility for any obligation whatsoever; and the fact that the government may have formulated, furnished, or in any way supplied the said drawings, specifications or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use or sell any patented invention that may in any way be related thereto.

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